

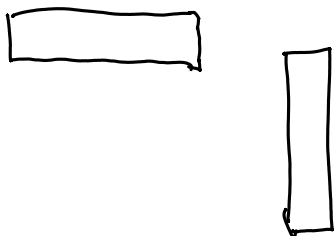
1.-

$$X \in [B, D], Y \in [D, F] \rightarrow Z \in [B, F]$$

1  
//

$$\bullet \text{ Bytes Per Int8} \cdot (B \cdot D + D \cdot F + B \cdot F)$$

$$= B \cdot D + D \cdot F + B \cdot F$$

-   $B \cdot F$  dot products
- Dot product has  $D$  multiplications

$$\text{and } a_1 \cdot b_1 + a_2 \cdot b_2 + \dots + a_D \cdot b_D$$

$D-1$  sums. Total OPs is then

$$\text{OPs} = B \cdot F \cdot (D + D - 1) = B \cdot F \cdot (2D - 1)$$

- $AI = \frac{B \cdot F \cdot (2D - 1)}{BD + DF + BF}$

- $T_{math} \stackrel{(*)}{=} \frac{B \cdot F \cdot (2D - 1)}{3.94 e^{14} / s}$

$$\begin{aligned} G &= 10^9 \\ T &= 10^{12} \end{aligned}$$

$$\stackrel{(*)}{=} \frac{B \cdot F \cdot (D - 1)}{2 e^{14}} \text{ seconds}$$

- $T_{comms} = \frac{B \cdot D + D \cdot F + B \cdot F}{8.2 e^{11} / s}$

- $\text{Max}(T_{comms}, T_{math}) \leq T_{operation} \leq T_{comms} + T_{math}.$

## Question 2:

- Accelerator FLOPs/s =  $1.97 e^{14} \simeq 2 e^{14}$   
for bf16:
- bf16 [B, D]<sub>D</sub> @ int8 [D, F]  $\rightarrow$  bf16 [B, F]
- Assuming no copy needed to cast int8 to bf16.
- $$A_i = \frac{B \cdot F \cdot (2D - 1)}{2B \cdot D + DF + 2 \cdot BF} \gg \frac{1.97 e^{14} \text{ FLOPs}}{8.2 e^{11} \text{ Bytes}}$$
$$\simeq \frac{2}{8} \cdot \frac{10^{14}}{10^{11}} = \frac{1}{4} \cdot 10^3 = 2.5 \cdot 10^2$$
$$= 250 \frac{\text{FLOPs}}{\text{Byte}}$$

Approximating  $A: \approx \frac{2 \cdot B \cdot F \cdot D}{DF}$

$$D \gg B$$

$$F \gg B$$

$$DF \gg 2 \cdot BD$$

$$DF \gg 2 \cdot BF$$

then  $DF \gg 2 \cdot BD + 2 \cdot BF$

$$\Rightarrow \frac{2 \cdot B \cdot F \cdot D}{DF} > 250$$

$$B > 125$$

"Per second I do more starts of algorithm since

less bytes to be transfer, so then per algo FLOPs can be smaller"

3:-

- Accelerator hfloat16  $1.47e14$
- Accelerator HBM  $8.2e11$
- Achievable FLOPs

$$= AI \times BW$$

$$\frac{2BD}{2BD + DF + 2BF} \cdot BW$$

$$D = F = 4096 \quad | \quad F = D = 1024$$

Achievable FLOPs/s

$$= \max \left( \frac{2BD^2}{2BD + D^2 + 2BF} \cdot BW, 1.97 \text{ e14} \right)$$

Assuming  $D = F$

$$\frac{2 \cdot B \cdot D^2}{2 \cdot B \cdot D + D^2 + 2 \cdot B \cdot D}$$

$$= \frac{2 \cdot B \cdot D^2}{4BD + D^2} = \frac{2 \cdot B \cdot D^2}{D \cdot (4 \cdot B + D)}$$

$$= \frac{2 \cdot B \cdot D}{4B + D} \quad D = 2^k$$
$$= \frac{2 \cdot B \cdot 2^k}{4 \cdot B + 2^k}$$

$$\frac{2 \cdot B \cdot 2^K}{}$$

$$2^K \cdot \left( \frac{1 \cdot B}{2^{K-2}} + 1 \right)$$

$$\frac{2 \cdot B}{\frac{1 \cdot B}{2^{K-2}} + 1} \cdot BW$$

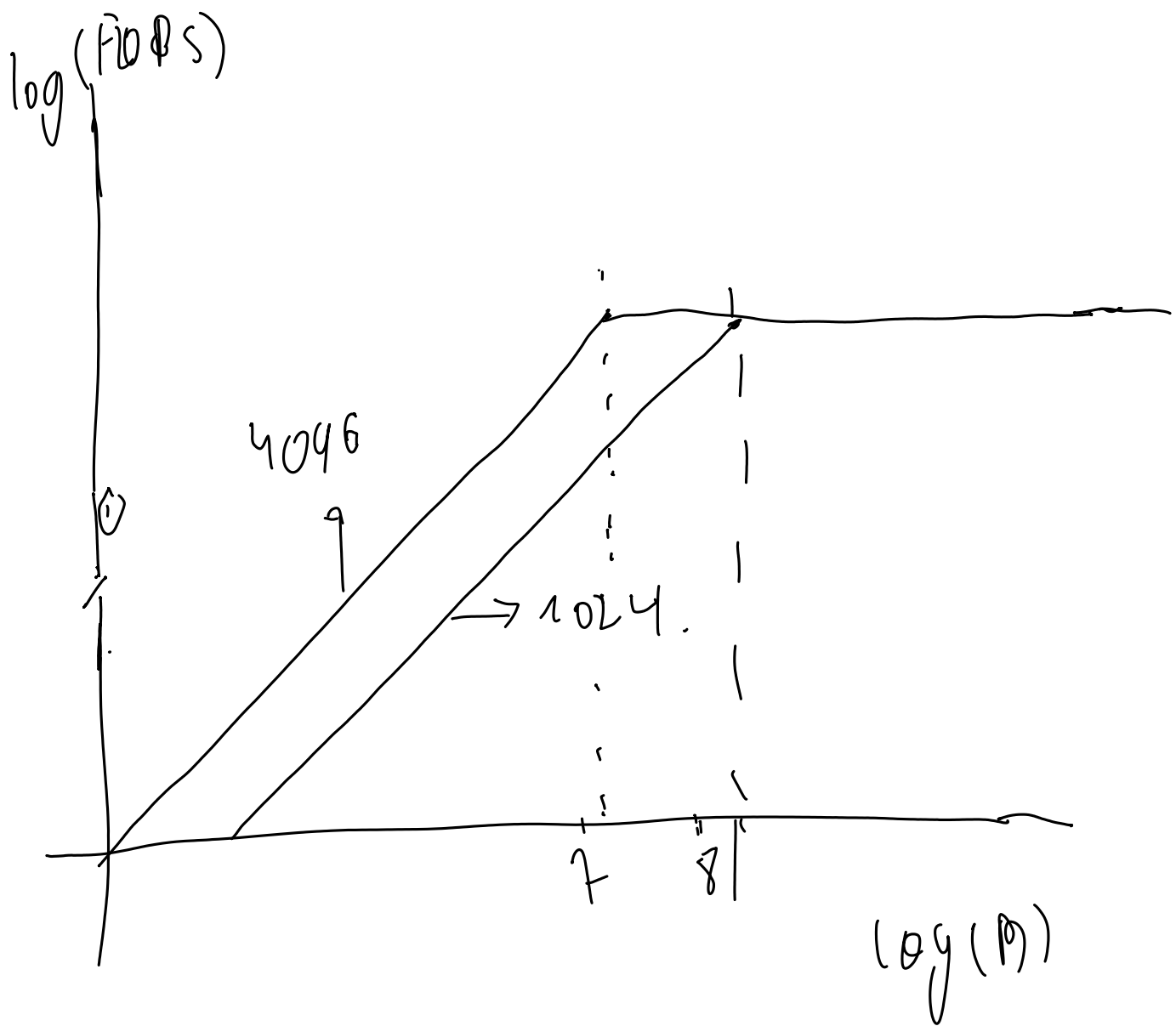
$$K = 12 \text{ for } N = 4096$$

$$K = 10 \text{ for } N = 1024$$

0

$$= \log(2 \cdot B) - \log\left(\frac{1 \cdot B}{2^{K-2}} + 1\right) + \log(BW)$$

$$\approx \log(2 \cdot B) - \frac{B}{2^{K-2}} + \log(BW)$$



$$\frac{2B \cdot 4096}{4B + 4096} \geq 250 \quad \frac{\text{FLOPs}}{\text{token}}$$

$$B \approx 142$$

$$\frac{2B \cdot 1024}{4B + 1024} \geq 250 \quad B \approx 244$$

4:-

B times  $\text{int8}[D] @ \text{int8}[D, F] \rightarrow \text{int8}[F]$

$$\text{FLOPS} = B \cdot (2FD)$$

$$\Rightarrow \text{AI} = \frac{B \cdot 2FD}{B \cdot (D + DF + F)}$$

$$= \frac{2FD}{D + DF + F}$$

5:-

$$\text{Accelerator FLOPS} \quad \frac{1974}{2} e^{12} \approx 984.5 e^{12}$$

$$= 9.89 e^{14}$$

$$\frac{2BF D}{2BD + 2DF + 2 \cdot BF} \Rightarrow \frac{4.84 \text{ e } 14}{3.75 \text{ e } 12}$$

$$\begin{array}{l} D \Rightarrow B \\ F \Rightarrow B \end{array} \Rightarrow \frac{\cancel{2BF D}}{\cancel{2DF}} \quad \begin{array}{l} 7/ \\ 295 \end{array}$$

$$B \quad 7/ \quad 295.$$

